

Access to Care in Rural America: The Experience of an Academic Medical Center in Mississippi

Sarah Bridgeman, MD
PGY-3 Dermatology Resident
University of Mississippi Medical Center



Disclosures

- I have no disclosures related to this talk

About Mississippi

- Rural State with 2.9 million citizens
- Land of contrasts
- The poorest state
- History tied to segregation and civil rights



Tips
e Sam Carr
ends





To understand the world, you
must first understand a place
like Mississippi

William Faulkner

Mississippi Data

- High rates of:
 - Diabetes
 - Heart Disease (The Jackson Heart Study)
 - Infant mortality
 - Uninsured

Objectives

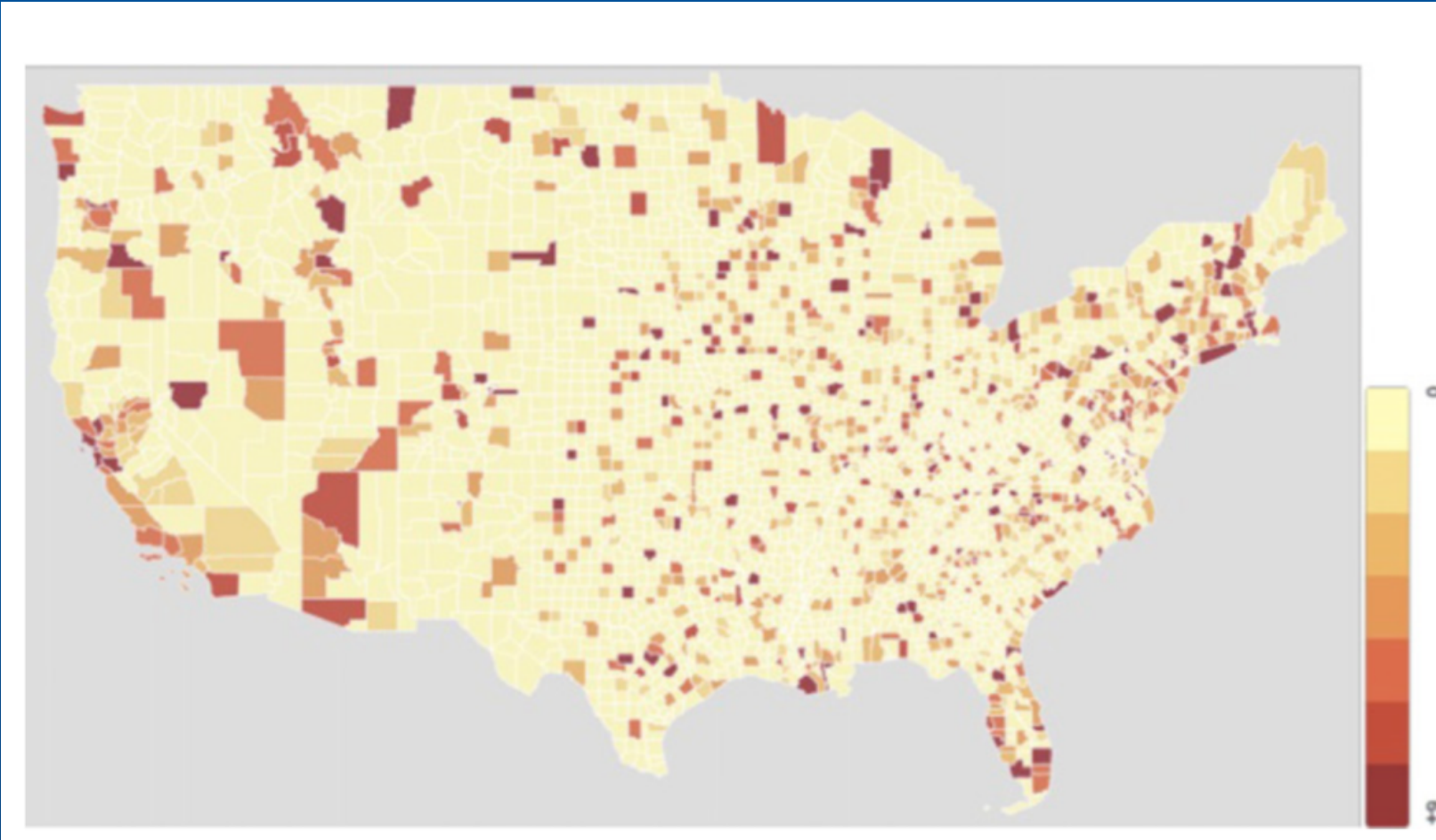
- The Premise
- The Problem
- A Plan with something for everyone
- A Promise
- The Pitfalls

The Premise

- Rural Areas in the US lack adequate Access to Dermatology Care (Dermatology Deserts)

The National Problem

County-specific dermatologist per population ratio distribution of the United States (per 100,000 persons)



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Vaidya, Toral et al.
Socioeconomic and
geographic barriers
to dermatology
care in urban and
rural US
populations. J Am
Acad Dermatol. 78
(2): 406 - 408.

6+

A National Problem

More Mid-level
care- givers serving
rural areas is not
the whole answer....

The Distribution of the U.S. Primary Care
Workforce. Content last reviewed July
2018. Agency for Healthcare Research
and Quality, Rockville, MD.

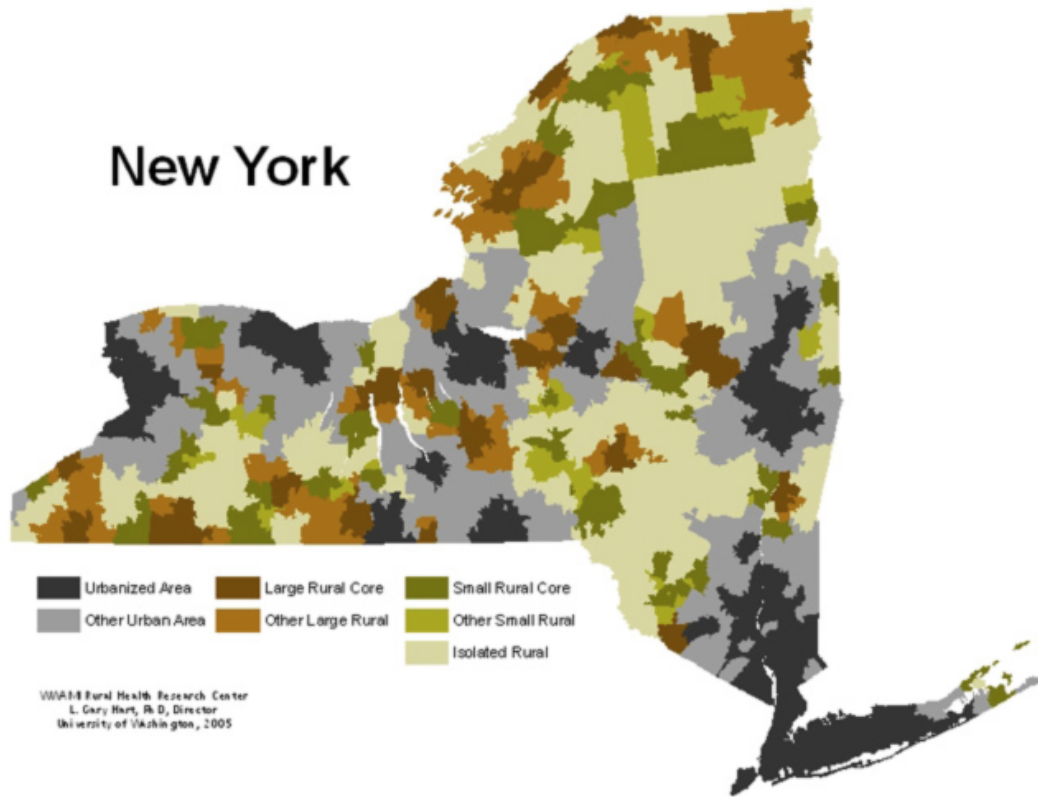
Geography	All specialties		
	NP	PA	Physicians
Urban	84.4%	84.4%	89.0%
Large rural	8.9%	8.8%	7.1%
Small rural	3.9%	3.8%	2.6%
Remote rural/frontier	2.8%	3.0%	1.3%

A National Problem

- 10-28% of Americans live in rural areas (approximately 30-90 million).^{1,2,3}

1. Hart LG, Larson EH, Lishner DM. Rural definitions for health policy and research. *Am J Public Health*. 2005;95(7):1149-1155.
2. Johnson-Webb KD, Baer LD, Gesler WM. What is rural? issues and considerations. *The Journal of Rural Health*. 1997;13(3):253-256.
3. Douthit N, Kiv S, Dwolatzky T, Biswas S. Exposing some important barriers to health care access in the rural USA. *Public Health*. 2015;129(6):611-620.

Maldistribution of Docs Is Real



Mississippi



Adapted from L Gary Hart. Rural Health Research Center.
University of Washington. 2005. Online.

Especially Dermatology in Mississippi

- 86 Dermatologists clustered in urban areas
 - 1/2 the per capita dermatologists as the average state
- When primary care doc in the Delta schedules a dermatology appointment for a patient
 - 3 month wait
 - 14% arrive at the office

Is Maldistribution A Problem?

- Rural patients more likely to die of preventable heart disease, stroke, lower respiratory tract disease & cancer

Richman L, Pearson J, Beasley C, Stanifer J. Addressing health inequalities in diverse, rural communities: An unmet need. *SSM-population health*. 2019;7:100398.

Cosby AG, McDoom-Echebiri MM, James W, Khandekar H, Brown W, Hanna HL. Growth and persistence of place-based mortality in the united states: The rural mortality penalty. *Am J Public Health*. 2019;109(1):155-162.

Does Maldistribution Affect Outcomes?

The association between dermatologist density and melanoma prognosis using melanoma mortality-incidence ratio

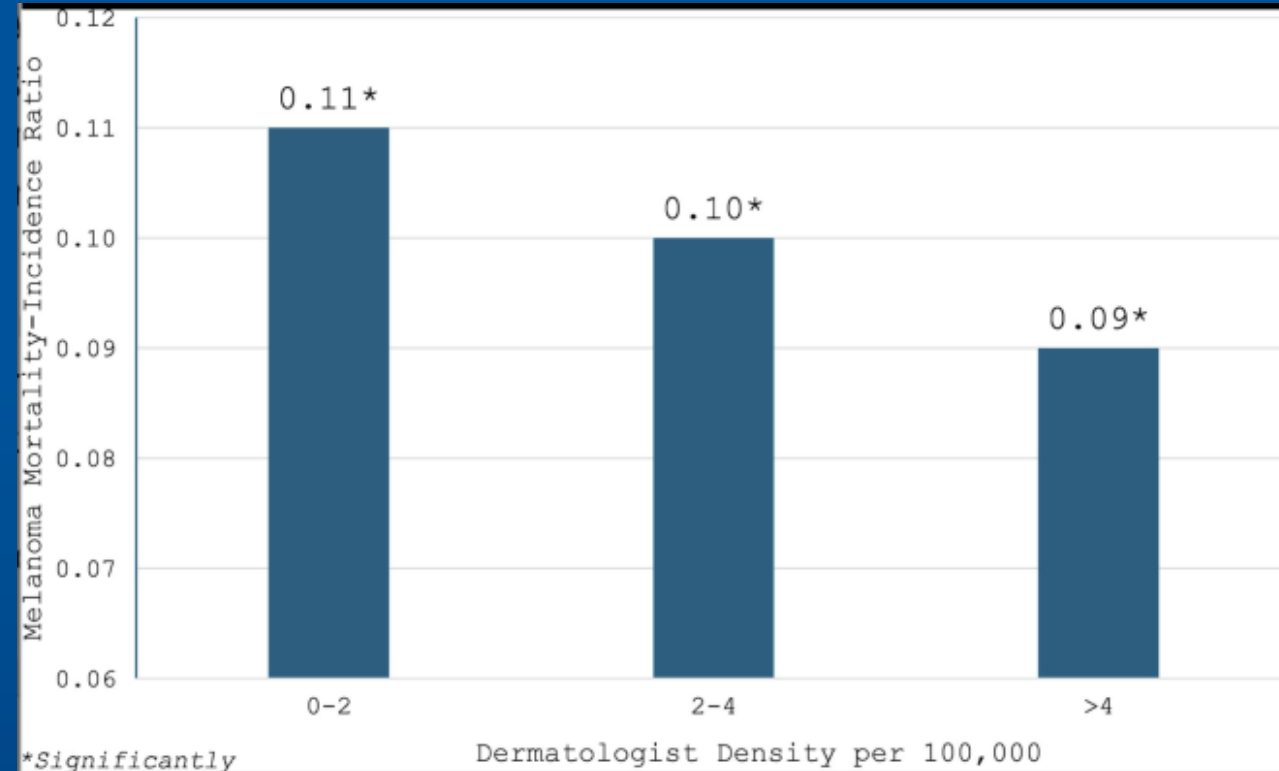
Burshtein, Joshua^a; Shah, Milaan^a; Zakria, Danny^a; Rigel, Darrell^b

Author Information

Melanoma Research 34(6):p 550-552, December 2024. | DOI: 10.1097/CMR.0000000000001004

Melanoma has the highest US mortality rate among skin cancers. The presence of more dermatologists has been associated with improved odds of early melanoma diagnosis [1]. Mortality-to-incidence ratio is the aggregate number of melanoma-specific deaths in a geographic region divided by the aggregate number of melanoma cases in that region (melanoma-incidence ratio [MIR] = melanoma deaths/cases). MIR is a better measure to assess the impact of dermatologist density on invasive melanoma prognosis than mortality alone, as it adjusts for regional differences in an absolute number of melanomas [2]. Dermatologist density in the USA is rising, though it remains below the recommended level for adequate population coverage (>4/100 000) [3]. An analysis of data from 1988 to 1993 found that higher dermatologist density was associated with lower melanoma MIR [2]. The purpose of this study was to quantitate at what levels dermatologist density positively

Mortality-to-incidence ratio =
melanoma deaths/cases



Burshtein, Joshua^a; Shah, Milaan^a; Zakria, Danny^a; Rigel, Darrell^b. The association between dermatologist density and melanoma prognosis using melanoma mortality-incidence ratio. *Melanoma Research* 34(6):p 550-552, December 2024. | DOI: 10.1097/CMR.0000000000001004

The Future is NOT Bright

Dermatology workforce projections in the United States, 2021 to 2036

Sarah Balboul ^{1 2}, Christian Gronbeck ², Hao Feng ³

Affiliations + expand

PMID: 38775980 DOI: [10.1007/s00403-024-02924-2](https://doi.org/10.1007/s00403-024-02924-2)

Abstract

Background: There has been a growing imbalance between supply of dermatologists and demand for dermatologic care. To best address physician shortages, it is important to delineate supply and demand patterns in the dermatologic workforce. The goal of this study was to explore dermatology supply and demand over time.

Methods: We conducted a cross-sectional analysis of workforce supply and demand projections for dermatologists from 2021 to 2036 using data from the Health Workforce Simulation Model from the National Center for Health Workforce Analysis. Estimates for total workforce supply and demand were summarized in aggregate and stratified by rurality. Scenarios with status quo demand and improved access were considered.

Results: Projected total supply showed a 12.45% increase by 2036. Total demand increased 12.70% by 2036 in the status quo scenario. In the improved access scenario, total supply was inadequate for total demand in any year, lagging by 28% in 2036. Metropolitan areas demonstrated a relative supply surplus up to 2036; nonmetropolitan areas had at least a 157% excess in demand throughout the study period. In 2021 adequacy was 108% and 39% adequacy for metropolitan and nonmetropolitan areas, respectively; these differences were projected to continue through 2036.

Conclusions: The findings suggest that the dermatology physician workforce is inadequate to meet the demand for dermatologic services in nonmetropolitan areas. Furthermore, improved access to dermatologic care would bolster demand and especially exacerbate workforce inadequacy in nonmetropolitan areas. Continued efforts are needed to address health inequities and ensure access to quality dermatologic care for all.

- Inadequate supply of dermatologists in any year through 2026
- Metro areas slight supply surplus → 2036
- Nonmetro areas had at least a 157% excess in demand

Balboul S, Gronbeck C, Feng H. Dermatology workforce projections in the United States, 2021 to 2036. Arch Dermatol Res. 2024 May 22;316(5):192. doi: [10.1007/s00403-024-02924-2](https://doi.org/10.1007/s00403-024-02924-2). PMID: 38775980

Before Devising A Plan:

What Do We Know About Making Rural Dermatologists?

1. Train More Residents

If urban areas become crowded with physicians, there will be a rural “trickle-down” effect

2. Choose “Special” Medical Students

Choose medical students from small towns

Train residents in rural areas

Expose trainees to rural medicine

Chan BT, Degani N, Crichton T, et al. Factors influencing family physicians to enter rural practice: does rural or urban background make a difference?. *Can Fam Physician*. 2005;51(9):1246-1247.

3. Build New Residency Program in Rural State, Rural Area of Your State, or one of 5 territories

Dermatology Residencies in the US (n=147)

11 states with no residency programs

AK, DE, HI, ID, ME, MT, NV, ND, SD, WY

Territories, only Puerto Rico has derm (2).

Who pays start-up costs?

Streifel, A, LL Wessman, RS. Farah, KJ Gaddis, A Byrd, RT Brodell, C Firkins Smith. Rural Residency Curricula: Potential Target for Improved Access to Care? *Cutis* (2021) 107 (1): 54-55

What Happened When a UMMC Dean Took the First Step and Committed to having a Dermatology Department?

In just 14 years, faculty recruited and graduates who stayed in Mississippi account for 40% of all dermatologists in the state

Current Statistics

- *50,000 patient visits per year with 20% from rural zip codes (10,000/year)*
- *Almost 2,000 Mohs cases/year*
- All subspecialty areas covered
- Train 5 residents per year

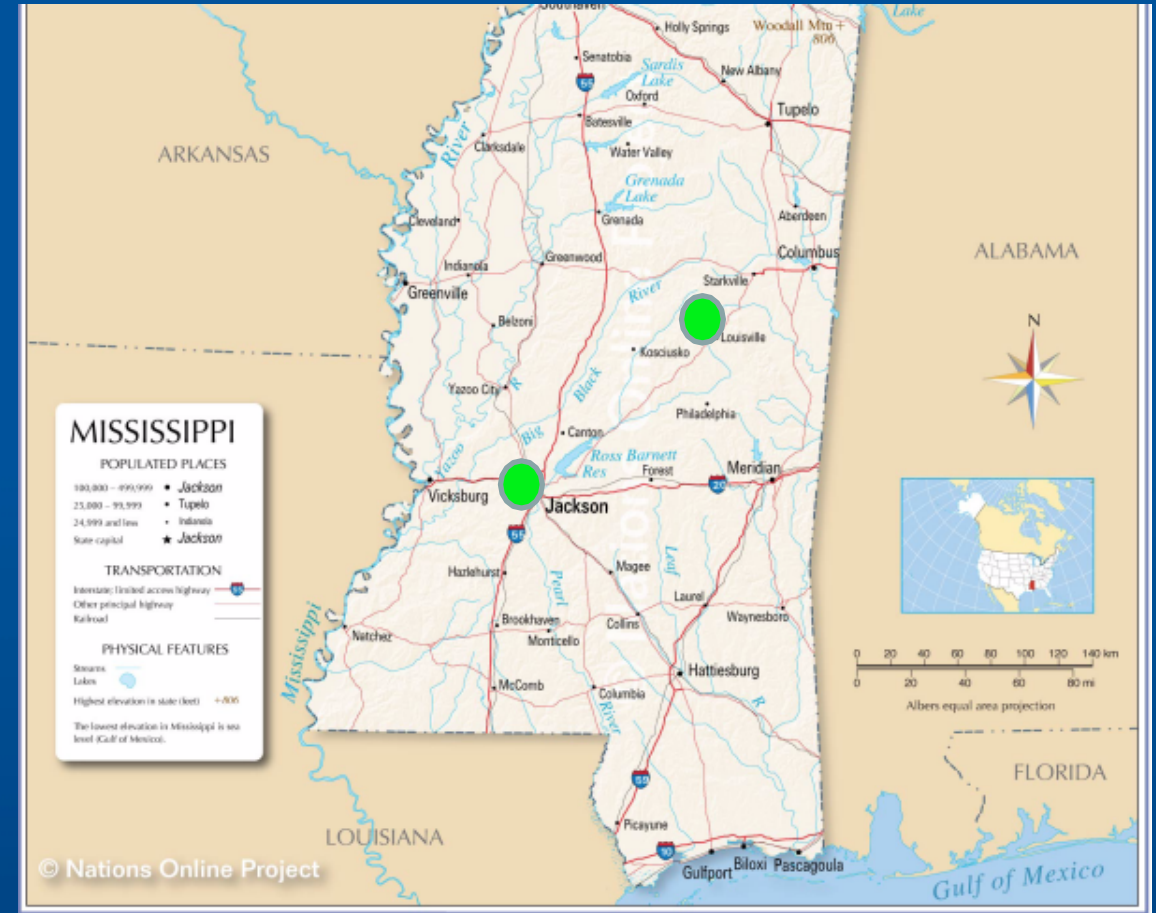
4. What Else Can Dermatology Departments Do??

A. Establish clinic space in rural areas

UMMC opened an academic office in Louisville, MS

Description

- Location: Louisville, MS: 90 miles from Medical Center
- Physician salary, benefits, incentive pay same as “mother ship”
- Dermatology residents, medical students, and FP residents rotate



How Accomplish That??

- Dermatologist with academic interest, desire to live in rural location (hometown roots)



Result: Improved Access

- Visits: 4500 visits per year
- RVUs: 75,000 RVUs/5 years
- Average wait time for new patient
 Louisville: 17-19 days
 Main campus dermatology: 34 days

Result: Profitability

- Dr Byrd at 100 percentile for Assoc Prof in academic practice
- Academic dermatology office which functions more like a private office

Result: Tertiary Referrals

Dermatopathology - 800/year

Mohs Surgery - 50/year

Cosmetic Dermatology

Complex Medical Dermatology

B. Establish a Rural Dermatology Track

The Essential Elements of Rural Track

- Reserve one rural residency slot annually
- 3 contiguous months rotations
- 3-year commitment at end of residency in rural area....probably their hometown!

Before You Say: Can't Fund!!

- UMMC - from rural hospitals and CMS 126
- St Louis University - Rural group dermatology practice pays rural resident and state grant
- University of Alabama - State Medical Society pays rural resident salary
- Penn State University-Generous donor and pharma
- Indiana State University
- George Washington University (Underserved)-Pharma
- University of Arkansas - Planning
- Ohio State University - Planning

UMMC received federally funded rural residency dermatology GME positions in 2022 (CMS 126- 200 rural slots per year for 5 years) - renewed in 2026

The CMS Rural Health Transformation (RHT) Program is a 50-billion dollar program with hundreds of millions of dollars provided to each state for high level initiatives to improve rural health. Money could trickle down...

HRSA (Health Resources and Services Administration) **Rural Residency Planning and Development (RRPD) Program**

Awards up to \$750,000 per grant.

What? How?

Supports start-up costs to create new rural residency programs & Rural Tracks. Funding for accreditation costs, faculty development, and resident recruitment. (both direct and indirect)

NOT currently for dermatology rural residencies

C. Retain Rural Residents in Rural Areas

80% of our rural residents are in rural America or contractually obligated.

D. Is Rural Resident Training Non-Inferior?

Fact:

- Great Education: 3 contiguous months in Louisville leads to more procedural opportunities, continuity, and many patients with rare/advanced disease

Fact:

- Comfortable Home: 3 nights per week in B&B - \$90/night paid by department (less than \$1,000/yr)
- Didactics: Monday in Jackson

Fact:

Quieter

Calmer

VERY grateful patients

Hunting, fishing, hiking,
biking





“Regular” Residents Who Stayed in Mississippi

Molly Webb, MD (UMMC)

Thy Huynh, MD (UMMC)

Chelsea Mockbee, MD (UMMC)

Allison Cruse, MD (UMMC)

Teel Liddell, MD (UMMC)

Ashton Davis, MD (Jackson Private Practice)

Caitlin Reed, MD (Jackson Private Practice)

Colleen Drapcho, MD (Jackson Private Practice)

Kenneth Saul, MD (Meridian)

Anna Wile, MD (Meridian)

Maurren Offiah, MD (Hattiesburg)

UMMC Dermatology

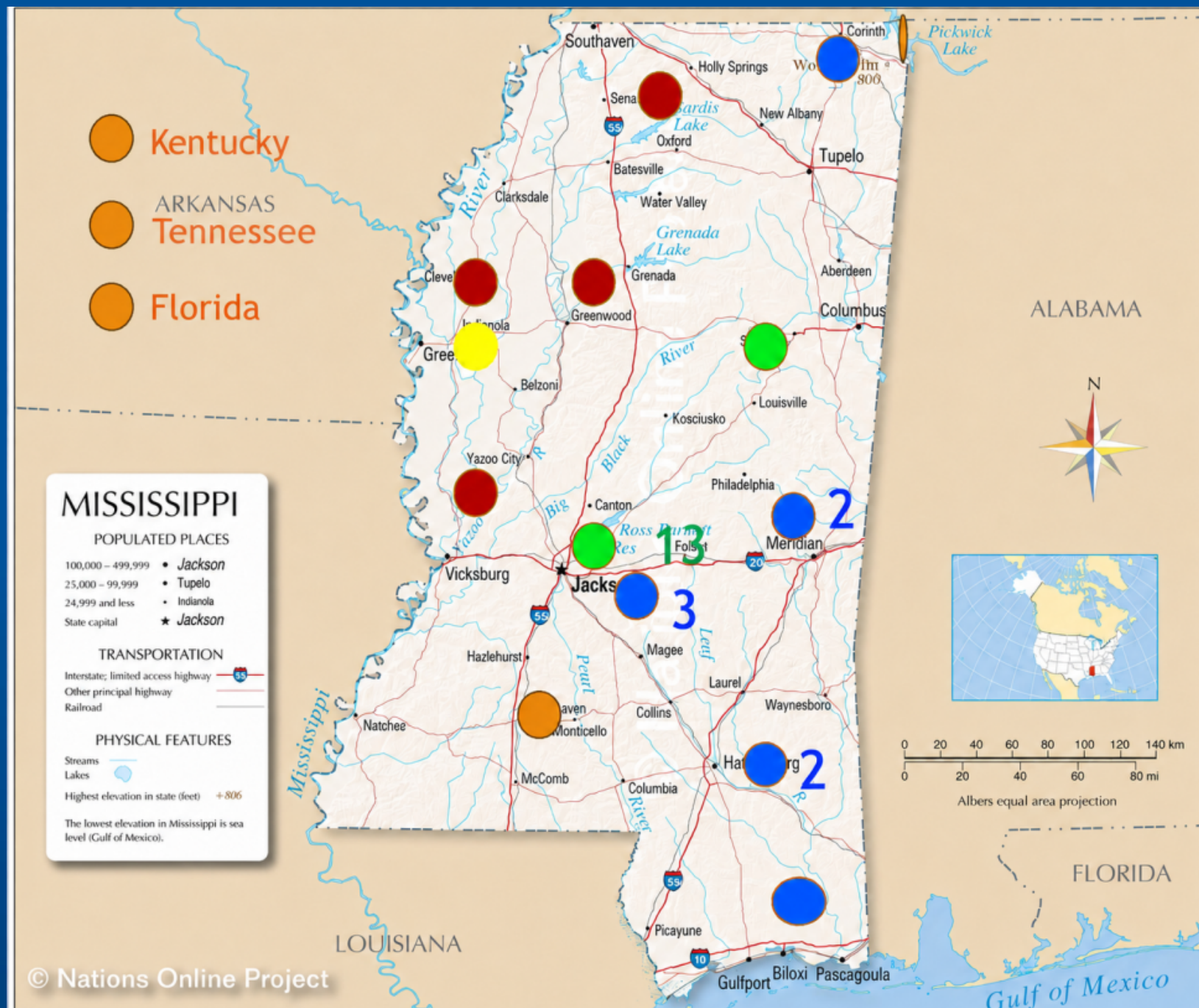
 UMMC Locations

 Private Graduates

 Committed Resident

 Free Clinic

 Future Locations



Pitfalls

- Ethical Issue: Handshake agreement for rural practice
- Exception to "all in" provision of MATCH
- Small pool of rural medical students
- Deployment to Kuwait

Comparative Effectiveness

What Really Works?

Effects over a decade....

	Patient visits/year	Patient visits/derm career
1 Rural Academic Office	5,000	200,000
8 of 10 rural residents honor commitment in decade	40,000	1,600,000
7 additional dermatology departments start program	280,000	11,200,000
TOTAL		13,000,000

Just Do It! Make
Access to Care a
Priority!

Never doubt that a small group of thoughtful, committed citizens can change the world; indeed, it's the only thing that ever has.

Margaret Mead



RURAL ACCESS TO
DERMATOLOGY SOCIETY



www.radsociety.org

